

Joshua Posner

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EDUCATION

Northeastern University

Expected: Apr 2029

B.S. Computer Science and Mathematics

Boston, MA

- **GPA: 3.95/4.0** (Dean's List 2x)
- **Relevant coursework:** Algorithms & Data, Program Design & Implementation I/II, Linear Algebra

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Rust, Python, Java, JavaScript, HTML, CSS
- **Frameworks:** OpenGL, React, GLFW, POSIX
- **Tools:** Git, CMake, Linux/Unix, VSCode

PROJECT EXPERIENCE

Onyx

- Identified a gap in accessible, high-level C++ rendering libraries; built a cross-platform rendering engine that abstracts low-level graphics APIs (OpenGL) behind a modern, RAII-based C++ interface.
- Reduced the code to render a directionally lit 3D model with live text overlay from 1,000+ lines of raw OpenGL/GLFW/Freetype/Assimp to ~80 easy-to-follow lines of Onyx API calls.
- Implemented core subsystems from scratch (windowing, text rendering, shader/pipeline management), applying object-oriented design and resource management patterns for a safe, leak-free API exceeding 25,000 lines of C++.
- Open-sourced the engine with Doxygen documentation, custom integration testing, and step-by-step tutorials; earned 50+ GitHub stars from developers seeking a lighter-weight alternative to existing engines.

Garnet

- Identified a similar lack of beginner-friendly C++ networking libraries; developed a cross-platform networking engine abstracting Windows/POSIX socket APIs into a unified, easier-to-use interface.
- Reduced the code for a multi-client TCP chat server with connection events and broadcast messaging from ~700 lines of raw Windows/POSIX to ~120 lines of Garnet API calls.
- Implemented two abstraction layers (raw platform-specific sockets into cross-platform sockets into multi-protocol server/client classes) and eliminated platform-specific edge cases (blocking vs. non-blocking I/O, socket lifecycle differences).
- Open-sourced the project with Doxygen documentation and example programs; received 20+ GitHub stars.

Plow n' Plate

- Partnered with one collaborator to prototype and partially implement a farming/cooking hybrid game in Rust using the Bevy ECS engine; gained hands-on experience with a newer language/paradigm outside C++.

WORK/VOLUNTEER EXPERIENCE

Teaching Assistant

2024 – 2025

- Designed 10+ assignments covering OOP, recursion, and data structures; still in use today.
- Graded 100+ submissions and frequently mentored struggling students one-on-one.

Connecticut Veterinary Center

July 2024 – Present

- Served as a primary point of contact for hundreds of clients daily, managing scheduling, medication dispensing, and billing/insurance inquiries – frequently handling high volume under time pressure.
- Assisted veterinarians with clinical tasks (preparing cost estimates for clients, drawing up vaccines).
- Worked 40+ hrs/wk during summers; 1,400+ hours total.